

Reading and Writing Console I/O

Class	Reading
Scanner	Easy input of different data types
BufferedReader	Fast text/line input
Console	Direct console input, including hidden passwords

Reading Console Input using Scanner Class

```
import java.util.Scanner;
class Demo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = sc.nextLine();
        System.out.println("Hello " + name);
    }
}
```

Enter your name: Ashok
Hello Ashok

Reading Integer

```
import java.util.Scanner;
class Demo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        System.out.println("Number = " + n);
    }
}
```

Reading Two Numbers and Adding

```
import java.util.Scanner;
class Demo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        int sum = a + b;

        System.out.println("Sum = " + sum);
    }
}
```

Enter first number: 10
Enter second number: 20
Sum = 30

Reading Different Data Types

```
import java.util.Scanner;

class Demo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter name: ");
        String name = sc.nextLine();

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        System.out.print("Enter salary: ");
        double salary = sc.nextDouble();

        System.out.println("Name = " + name);
        System.out.println("Age = " + age);
        System.out.println("Salary = " + salary);
    }
}
```

Method	Reads
nextInt()	Integer
nextFloat()	Float
nextDouble()	Double
next()	One word
nextLine()	Complete line
nextBoolean()	true / false

Reading Console Input Using BufferedReader

BufferedReader can be used to read input from the keyboard (console).
It reads text as characters/lines.

```
import java.io.BufferedReader;
import java.io.InputStreamReader;

class Demo {
    public static void main(String[] args) {

        try {
            BufferedReader br =
                new BufferedReader(new InputStreamReader(System.in));

            System.out.print("Enter your name: ");
            String name = br.readLine();

            System.out.println("Hello " + name);

        } catch (Exception e) {
```

```
        System.out.println("Error");
    }
}

import java.io.BufferedReader;
import java.io.InputStreamReader;

class Demo {
    public static void main(String[] args) {

        try {
            BufferedReader br =
                new BufferedReader(new InputStreamReader(System.in));

            System.out.print("Enter age: ");

            int age = Integer.parseInt(br.readLine());

            System.out.println("Age = " + age);

        } catch (Exception e) {
            System.out.println("Error");
        }
    }
}
```

Scanner	BufferedReader
nextInt()	Integer.parseInt(readLine())
nextDouble()	Double.parseDouble(readLine())
nextLine()	readLine()
Easier for beginners	Faster and efficient for text input

Reading Console Input Using Console Class

Java provides the Console class to read input directly from the console.

```
import java.io.Console;

class Demo {
    public static void main(String[] args) {

        Console c = System.console();

        System.out.print("Enter your name: ");
        String name = c.readLine();

        System.out.println("Hello " + name);
    }
}

Enter your name: Ashok
Hello Ashok

Reading an Integer
import java.io.Console;
```

```

class Demo {
    public static void main(String[] args) {

        Console c = System.console();

        System.out.print("Enter age: ");
        int age = Integer.parseInt(c.readLine());

        System.out.println("Age = " + age);
    }
}

```

Reading Password

```

import java.io.Console;

class Demo {
    public static void main(String[] args) {
        Console c = System.console();
        System.out.print("Enter password: ");
        char[] password = c.readPassword(); // nothing appears on screen at all
        System.out.println("Password entered");
    }
}

```

Writing to Console

```

class Demo {
    public static void main(String[] args) {
        System.out.println("Hello Java");
        System.out.println("Welcome to Programming");
    }
}

```

Hello Java

Welcome to Programming

Writing console output using write() method

```

class Demo {
    public static void main(String[] args) {

        System.out.write('A');
        System.out.write('B');
        System.out.write('C');
    }
}

```

ABC

Byte Stream

Reads just one single byte from input and writes just one single byte back out — nothing more, even if you type multiple characters.

```
import java.io.*;

class Demo {
    public static void main(String[] args) throws Exception {

        int data = System.in.read(); // Read one byte from console
        System.out.write(data);      // Write one byte in console
    }
}
Input: A
Output: A
```

Character Stream

Read and write characters

```
import java.io.*;

class Demo {
    public static void main(String[] args) throws Exception {

        InputStreamReader in = new InputStreamReader(System.in);
        OutputStreamWriter out = new OutputStreamWriter(System.out);

        int ch;

        while ((ch = in.read()) != -1) {
            out.write(ch);
            out.flush();
        }
    }
}
```